



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

June 16, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 13182
(SIXTH REVISION)

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: (See individual authorization letter)
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the transportation in commerce of phosphine in certain DOT specification and non-DOT specification cylinders. This special permit provides no relief from the Hazardous Materials Regulation (HMR) other than as specifically stated herein.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use or other uses not associated with transportation in commerce.
 - c. Unless otherwise stated herein, this special permit consists of the special permit authorization letter issued to the grantee together with this document.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.192(a) in that the DOT specification cylinders may exceed 125 lb. water capacity and that non-DOT specification cylinders are not authorized except as specified herein; § 171.23(a)(5), 171.23(a)(6), and 173.304(b) in that the prescribed DOT and non-DOT specification cylinders authorized herein are to be charged for import and export; and the cylinders may be liquid full with phosphine at temperatures below 130°F.
5. BASIS: This special permit is based on the application of Cytec Industries Inc. dated July 20, 2005, submitted in accordance with § 107.105 and the public proceeding thereon.

Tracking Number: 2024115303

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Proper Shipping Name/ Hazardous Materials Description	Hazard Class/ Division	Identi- fication Number	Packing Group
Phosphine	2.3	UN2199	N/A Zone A

7. SAFETY CONTROL MEASURES:a. PACKAGING: Packagings prescribed are:

- (1) DOT specification 3A, 3AA and 3AL cylinders with a minimum marked service pressure of 1800 psig (124 bar).
- (2) Non-DOT specification seamless aluminum cylinders with a marked test pressure of 300 bar (4,500 psig) manufactured by Luxfer Gas Cylinders (Nottingham, England). Cylinders must be in conformance with Luxfer Gas Cylinders drawing numbers NM20EEC, NM40EEC, and NM50EEC on file with the Office of Hazardous Materials Safety. The cylinders must be tested and re-qualified by a DOT approved facility and in accordance with the ultrasonic examination (UE) method described in latest revision of ISO 6406.

b. OPERATIONAL CONTROLS:

- (1) The water capacity of each cylinder is limited to a maximum of 57 Kg (125 lbs).
- (2) The aluminum cylinders may only be offered for transportation or transported by highway and rail.
- (3) The maximum permitted fill density is 37 percent. Fill density is defined as the percent ratio of the weight of the gas in a cylinder to the weight of the water that the cylinder will hold at 60°F.
- (4) The pressure in the cylinder may not exceed 95 percent of the cylinders test pressure at 65°C (149°F).
- (5) The critical temperature (T_c) of phosphine must be above 15°C (59°F) and below 65°C (149°F).
- (6) Pressure relief devices are prohibited.

(7) The shipping paper must identify the cylinders with the special permit number “DOT-SP 13182”. Additionally, when offered for export, the shipping papers must carry the following certification: “This cylinder has (These cylinders have) been retested and refilled in accordance with DOT requirements for export and the terms of this special permit.”

(8) Emergency response information provided with the shipment and available via an emergency response telephone number must indicate that the cylinders are not fitted with pressure relief devices and provide appropriate guidance for exposure to fire.

(9) The DOT specification cylinders authorized by paragraph 7.a.(1) of this special permit may be charged for import or export and also transported within the US; however, the non-DOT specification cylinders authorized by paragraph 7.a.(2) may only be charged for import or export.

8. SPECIAL PROVISIONS:

a. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

b. Each cylinder must be plainly and durably marked “DOT-SP 13182” as specified in § 172.301(c).

c. Packagings permanently marked “DOT-E 13182”, prior to October 1, 2006 may continue to be used under this special permit for the remaining service life of the packaging or until the special permit is no longer valid. Packagings marked on or after October 1, 2006, must be marked “DOT-SP 13182”.

d. Shipping papers displaying “DOT-E 13182” may continue to be used until October 1, 2006, provided the special permit remains valid.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle and cargo vessel.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each motor vehicle or cargo vessel used to transport packages covered by this special permit.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) --- “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR § 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Quade
Acting Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: BB